1 · Independent states

Question: does a larger sample remove measurement bias?

Answer: more observations reduce sampling variation; systematic measurement bias can remain.

Use comma/period to switch tiers without changing slides.

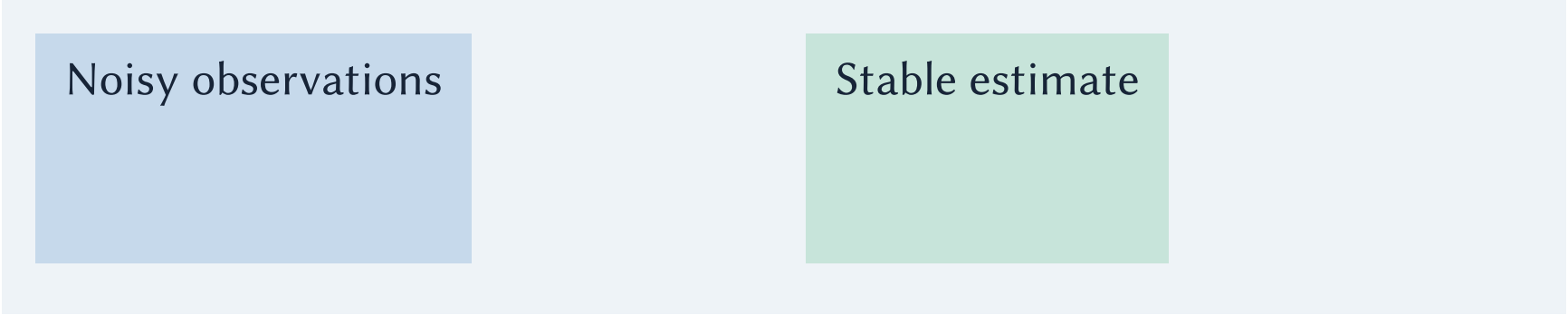
2 · A decorated table

Which estimator is more stable?

Estimator	Bias	Variance
Unregularized	0	9
Regularized	1	2

These are illustrative squared-error components, not empirical measurements. Lower variance alone does not establish lower total error.

3 · Figure and nested support



Stability depends on how the observations constrain the estimate.

4 · Multiple takeaways and a detour

More data need not remove systematic bias.

Regularization can trade bias for lower variance.

Enter follows the derivation detour. R restores this slide and its saved tier.

5 · Same-position opaque replacement

Correction: compare total error, including squared bias.

Opaque alternatives follow Typst paint order. The amber cover is on top in the complete PDF.

6 · Derivation detour

$$\mathrm{MSE}\left(\hat{\theta}\right) = \mathrm{Var}\left(\hat{\theta}\right) + \mathrm{Bias}\left(\hat{\theta}\right)^2$$

This page selects the flat preset: one complete state.

Press R after arriving through Enter to return to the exact saved tier.